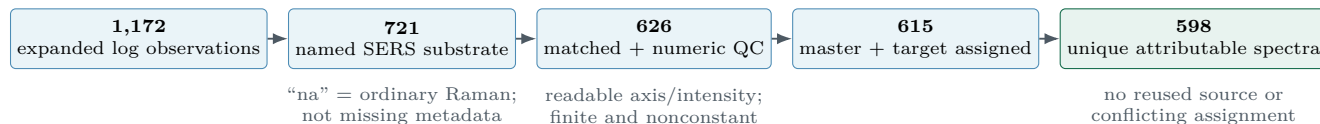


A source-reversible audit reduced **721 explicitly named SERS observations** to **598 unique, numeric, chemically attributable spectra** from **69 physical samples**. Repeated multi-instrument measurements make the dataset suitable for chemical identification under acquisition shift; they do not establish a clean-spectrum reference or causal nuisance separation.

1. Dataset construction



Resulting analysis population

- 598 spectra, but only **69 independent physical samples**.
- 10 instruments, 4 SERS sensor families, 3 stations, and 7 recorded labels.
- Repeated views remain linked by one physical-master identifier.
- The 500-spectrum notes-clear and 575-spectrum Mira-1-excluded tiers are retained for sensitivity analysis.

Primary spectral representation

- Common measured support: **400–1,800 cm^{-1}** ; no low-shift extrapolation.
- Interpolation at 1 cm^{-1} gives **1,401 ordered coordinates**.
- Each spectrum is independently min-max scaled to $[0, 1]$.
- SG and arPLS are retained as candidates; minimal scaling is the reference, not a claimed optimum.

Robust instrument spectral summaries

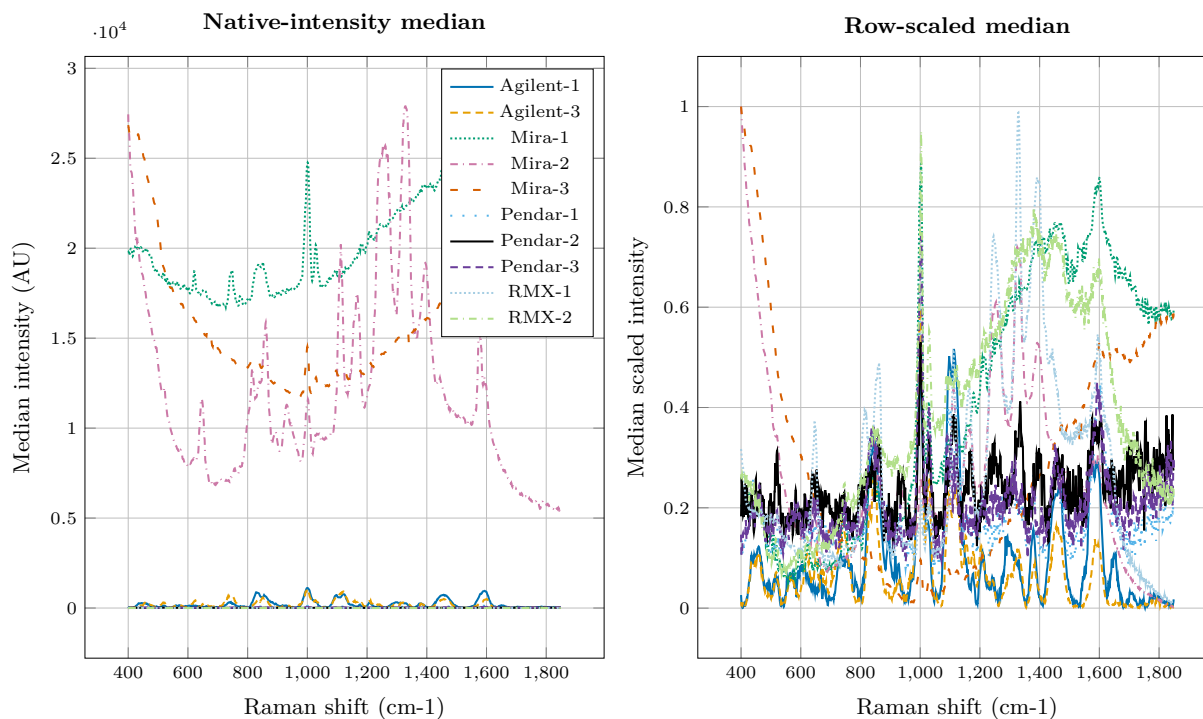


Figure 1: Instrument median spectra before and after row-wise scaling. Scaling removes the absolute intensity range but preserves visible acquisition-dependent background and shape differences.

2. Preliminary evidence and research direction

Acquisition dominates the raw geometry. Raw channel-standardized PCA assigns **93.68%** of variance to PC1. Row-wise min-max scaling reduces PC1 to **42.68%**; the remaining clusters are still more associated with instrument than chemistry.

Chemical information remains measurable. Combining repeated measurements within each physical sample increases chemical cluster association to **0.668**, although station association remains **0.585**. Chemistry and acquisition context are therefore not fully separated.

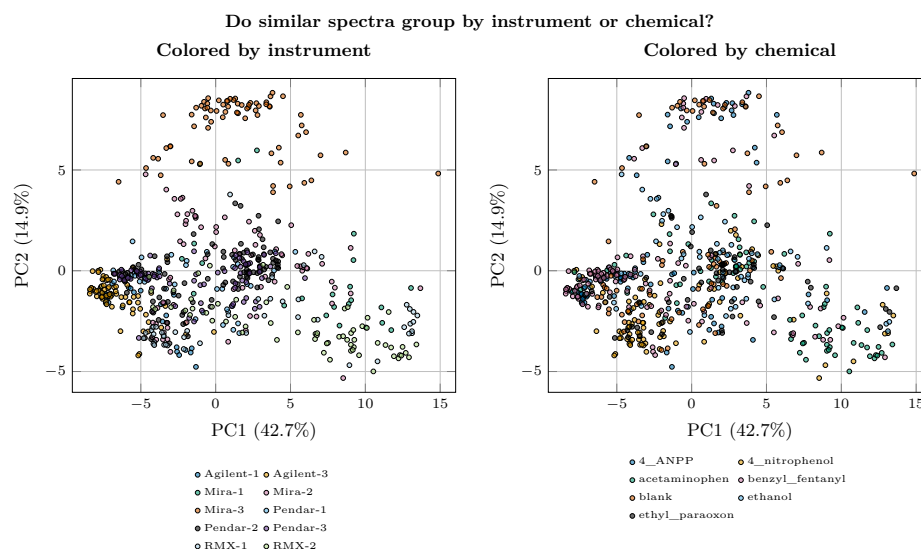


Figure 2: The same min-max PCA coordinates colored by instrument and chemical. Instrument labels produce clearer groups; PCA is descriptive and is not classification performance.

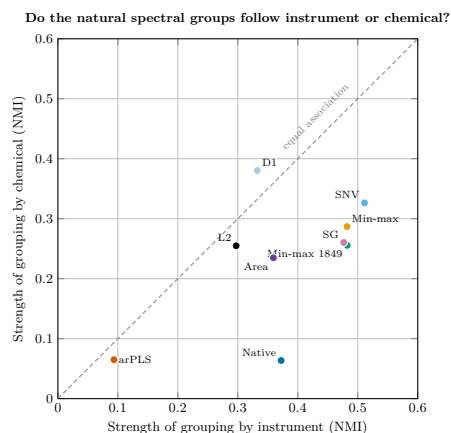


Figure 3: Cluster association under candidate representations. Below the diagonal indicates stronger instrument association.

Preprocessing trade-off. Min-max cluster association is **0.482** with instrument and **0.287** with chemical identity. arPLS lowers these values to **0.094** and **0.065**. Reduced instrument association alone therefore does not show that chemistry was preserved.

Questions supported by the data

1. Can supported chemicals be identified on an unseen instrument and unseen physical samples?
2. Do universal or source-selected preprocessing rules improve transfer without damaging peaks?
3. Do compact or acquisition-aware deep models improve on tuned classical methods?
4. What is gained from calibration samples, and can unfamiliar cases be flagged?

Why these questions fit. Repeated cross-instrument views expose acquisition shift while retaining physical-sample identity. The 69-master sample size favours controlled classical models and compact neural networks.

Current step—classical ML has started. Physical samples remain grouped, and one complete instrument is excluded from development. Spectral matching, PCA-LDA, PLS-DA, logistic regression, RBF SVM, Random Forest, and ExtraTrees establish the reference; compact and acquisition-aware deep models will use the same roles.

Boundary: no universally best preprocessing, clean-spectrum recovery, causal disentanglement, or arbitrary-domain generalization is claimed.